TERMINI · AI BUILDER CUP · 5-DAY CLOUD-VM DEMO

Termini AI-Infra

Turning bare, on-premise hardware into a **self-operating AI fleet** — a cloud control plane bootstraps it, a **governed AI operator** runs it, and **your data never leaves**.

“Your hardware. Your data. Our operations.”

Day-0 bootstrap | Day-2 self-healing | Fail-closed governance

AI is stuck at the on-prem gate

Enterprises holding governance-sensitive data — customer records, source code, national-core IP — **cannot ship that data to a SaaS cloud** to get AI. They are stuck at three points no vendor has productized into one thing.

1 Day-0 — turn it on

Standing up an AI-ready cluster (hypervisor, GPU serving, networking, agent VMs) today requires a specialist. A small or regulated business has **no SRE bench** to do it.

2 Day-2 — keep it alive

Once it runs, someone must watch disks, memory and GPU health and fix failures. On-prem means **you are the operations team**.

3 Governance — prove it

“Your data never leaves” has to be an **enforceable property**, not a slogan in the sales deck.

Ops can be delegated. Data doesn’t have to follow.

A Datadog proved delegation

Best-in-class day-2 observability plus an agentic surface. But it is **SaaS-architected** — telemetry, traces and logs ship to *its* cloud, with no self-hosted / air-gapped AI offering. And it assumes the infrastructure **already exists**.

B Samsung × Mistral proved demand

The Sept 2026 partnership keeps AI on-prem for data sovereignty. But it is **bespoke, equity-backed**, mega-corp — and covers **models & platform, not day-2 operations**. Small businesses can’t replicate it.

The **missing middle**: nobody productizes the **bootstrap** + the **day-2 operator** + the **governance boundary** in one thing.

“Datadog proved ops can be delegated — but it does it by moving your telemetry to their cloud. We do the same job for people **who can’t send anything off-prem**.”

Bootstrap the hardware. Operate it with a governed AI.

A cloud control plane bootstraps the hardware on day-0, an AI operator remediates it on day-2, and a fail-closed governance boundary makes data residency **checkable — not asserted**.

Telemetry → **Threshold rule** → **Advisory model** → **Deterministic clamp** → **Command allowlist** → **Audited action**

+ We build two things

The **governance boundary** and the **deterministic AI-operator control loop** — the moat.

= We adopt the rest

The substrate (LXD + Ubuntu), the connectivity (underlay + SSH CA), the model (llama.cpp + GGUF). Model choice is a **config knob** — the product is the ops + governance layer.

Same delegation. Opposite data flow.

vs Datadog (infra ops)

- Datadog **exfiltrates** telemetry to its SaaS; Termini runs a **self-hosted agentic remediation loop** — model on the customer’s own GPU.
- Datadog has **no self-hosted / air-gapped AI**; Termini serves an entire market that can’t send anything off-prem.
- Datadog assumes the infra **already exists**; Termini owns day-0 bootstrap too.
- Only **C0 hardware/OS telemetry** ever leaves Termini’s boundary.

vs Samsung × Mistral (on-site AI)

- Bespoke, equity-backed, **mega-corp** engagement — not replicable by an SMB.
- Covers **models and platform**, not maintenance and operations.
- Termini **productizes the long tail**: open-weight self-hosted models **plus** the bootstrap and the governed operator.
- Samsung has expert engineers; **SMBs do not** — that’s the wedge.

Three proof points

1 Bare metal → AI-ready fleet

A stock Ubuntu node boots the pre-baked image, registers to the controller in **under 5 seconds**, and joins the LXD cluster with its role assigned — **no SRE, no manual config**.

2 NL intent, guardrail-enforced

Type *“deploy 3 worker agents and a fast LLM for coding.”* An on-prem model extracts a structured allocation; the **deterministic guardrail clamps** the impossible and the allowlist turns it into **real VMs**.

3 Self-healing with a boundary

Kill a disk or OOM a VM live. Telemetry crosses a threshold, the operator **detects → decides → executes → audits** — while a C2 field is **blocked before any byte leaves**.

Q1 What operations scenarios do we showcase, and how do we simulate them?

RESOLVED

Two real failures, two real fixes — A: out-of-memory and B: disk degradation — simulated with off-the-shelf tools, so a judge sees the exact failure a production customer would.

A Out-of-memory

stress-ng burns RAM, starves neighbor agents → operator detects, **resizes** the offending VM, restores the fleet.

B Disk degradation

Injected **I/O error** (write latency climb / read-only mount) → operator **flags the drive, drains the node, alerts** the user.

Contested: beating the kernel OOM-killer is best-effort. Resolved with **1000 ms telemetry** + a **local fast-path** (no model latency) + a **synthetic persistent trigger** — we demonstrate the **detect → decide → execute → audit** loop **deterministically**.

HOW THE DEMO PROVES IT — dashboard shows RAM crossing 95% → a **vm.resize** acks **succeeded** → new ceiling visible, traceable end-to-end in the audit log.

Q2 What cloud system ties the nodes together — LXD or OpenStack — and how do they share a network?

RESOLVED

LXD (5.21/stable) as a 3-member cluster. OpenStack rejected — hours-to-days of setup for problems this demo doesn't have.

Networking — no overlay

Each node keeps a **local NAT bridge** on its own subnet; cross-node traffic rides the **physical underlay by real IP**. GPU inference is reachable at **http://10.10.0.13:8080**.

No tunnel to misconfigure

A single static route either works or it doesn't — **no encapsulation**. Guest MTU clamped to the underlay (**1460**) to avoid PMTU black holes.

Contested: overlay tunnels (OVN/Fan/VXLAN) silently drop packets on MTU mismatch and add per-call latency. Resolved by **removing the tunnel** entirely.

HOW THE DEMO PROVES IT — from an agent VM on Node A, **curl http://10.10.0.13:8080/v1/models** returns the model list over standard HTTP; watch **traceroute** and streamed tokens with no tunnel in the path.

Q3 How do we smartly allocate agent-VM resources — can we infer them from natural language?

RESOLVED

Yes. Plain English → intent extraction (on-prem, grammar-constrained JSON) → deterministic validator. The model recommends; the validator decides.

✓ Hard caps

- **vcpus ≤ 128**, **ram ≤ 512 GB**, **count ≤ 64**
- Per-node ceilings with **host reserve**
- Per-type ranges (agent_vm: vcpus 1–8, ram 1–16 GB)

✕ All-or-nothing

- Numbers are **clamped** to real capacity
- Structural violations **rejected** outright
- A rejected class creates **nothing**

Contested: a prompt like “super fast agent” yielding **vcpus:128, ram:512** that LXD can’t honor. Resolved with **deterministic hard guardrails, not model politeness**.

HOW THE DEMO PROVES IT — type **“128 vCPU, 512 GB RAM worker”** → the plan shows **clamped values** within the host’s real capacity, with per-field **from → to** clamping visible in the audit record.

Q4 How do we serve the pre-baked agentic OS — what's pre-installed, and how do we hand it over AI-ready (SSH-able)?

RESOLVED

Immutable pre-baked images — nothing downloads at boot. cloud-init only activates & registers in <5s. Access is browser-first.

◈ Access without passwords

- Browser generates the **keypair locally**
- Controller signs a **short-lived cert (≤8h)**
- One-click **web terminal** + an **ssh** string
- **No shared private key** — one leak can't compromise the fleet

▣ Data boundary (C0/C1/C2)

- Only **C0 hardware/OS telemetry** + **C1 transformed IDs** egress
- **C2** (logs, payloads) **never leaves**
- On-node redactor **fails closed** before the socket opens
- External model calls scoped to **C0 only**

HOW THE DEMO PROVES IT — power on → **registered <5s** with zero installs; open the terminal for a **passwordless shell**; pull the governance audit and show **zero C2 records**, every call traceable by its **call_id**.

10 LIVE DEMO

What you are about to see

A live, on-premise cluster running on bare hardware. Four moments, in order:

1 Node powers on → registered in <5s
Pre-baked image activates, joins the LXD cluster, role assigned. No package installs.

2 Plain English → real VMs
“Deploy 3 worker agents and a fast LLM for coding.” Spec extracted on-prem, clamped, executed.

3 OOM a VM / kill a disk — live
The AI operator detects, decides, executes the fix, and writes the audit trail.

4 Zero C2 data leaves
Watch the governance audit: only C0/C1 egress, the redactor holds the line.

▸ Narration takes over now — over to the live cluster.

11 RESULTS

What we just demonstrated

<5s

Bare-metal registration, **zero package installs**

clamped

Natural-language intent → **valid allocation** → real VMs, from→to visible

audited

Deterministic OOM & disk remediation, **end-to-end traceable**

0

C2 egress — the boundary held on every call

2

Fixes executed autonomously (resize, drain) from threshold rules

3

Proof points shown live, not claimed

What’s next after the 5-day build

- **Learned baselines**
  Netdata edge-ML anomaly detection replaces static thresholds.
- **HA shared storage**
  MicroCloud / MicroCeph for shared storage, multi-tenant isolation, no migration copy.
- **Model-agnostic serving**
  Swap DeepSeek distills for any open-weight model — the product is ops + governance, not the weights.
- **Scale & retention**
  VictoriaMetrics for persistence; k3s when agent scheduling moves to Kubernetes.
- **Real bare-metal onboarding**
  MAAS + iPXE netboot to onboard customer hardware end-to-end.
- **Compliance alignment**
  GDPR, EU AI Act, NIS2, DORA posture mapped to the governance boundary.

TERMINI · AI-INFRA

Your hardware.
Your data.
Our operations.

Bootstrap day-0. Self-heal day-2. Prove the data stays.

Questions

Live demo available